

Anubha Parashar

Analytics & AI Engineer

GenAI | LLM | NLP | Computer Vision | Deep Learning Artificial Intelligence | Machine Learning | IoT



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WORK EXPERIENCE

CURRENT, FROM JAN 2025

1. Pearce Services, Gurugram, India Analytical & AI Engineer

- Hybrid Optimization & Predictive Scheduling for Wind Turbine Maintenance:** Designed and implemented a hybrid scheduling algorithm integrating Ant Colony Optimization with Nesterov's accelerated gradient method, reducing technician travel distance by 25% and boosting schedule feasibility by 30%.
Developed a GRU-LSTM ensemble in TensorFlow/Keras to forecast 8-hour wind-speed trajectories (0.4 m/s MAE, 90% accuracy) using real-time SCADA and meteorological feature extraction with hyperparameter tuning, improving on-schedule maintenance by 25%, cutting idle time by 15%, and projecting \$200 K in annual cost savings.
- Solar Defect Detection with YOLOv11:** Developed and trained a YOLOv11 model on a 10,000+ image dataset (RGB + thermal) with multi-class annotations and extensive augmentation, achieving 0.92 mAP and 0.88 recall for detecting cracks, diode failures, and hotspots, etc. and deployed it to NVIDIA Jetson edge devices for real-time inspections, reducing manual survey time by 70%.
- Offline Mobile Chatbot for Maintenance Support:** Implemented on-device semantic retrieval using DeepSeek embeddings and FAISS indexing, enabling offline query responses, cutting manual content updates by 50%, and improving answer accuracy by 30%.

APR 2024 – DEC 2024

2. Invincible Ocean, Gurugram, India Research Scientist - AI and Metaverse

- Vehicle RC Data Chatbot:** Fine-tuned a Large Language Model to handle vehicle RC data queries. Developed a user-friendly chatbot, reducing manual search time by 70% and providing instant access to vehicle information.
- Virtual Try-On (VITON) System:** Created a Virtual Try-On platform using deep learning to overlay clothing on user photos, achieving a 95% user satisfaction rate.
- Text Recognition and OCR System:** Designed and trained neural networks with Tesseract, PaddleOCR, and Paddle NLP, attaining 98% accuracy. Developed a Label Studio backend for auto-annotation, APIs for OCR on base64 images, Dockerized various components, and created a Gemini assistant for information extraction.

JAN 2018 – APR 2023

3. Doctoral Research Robust Gait Recognition System Using Deep Learning to Handle Co-variables

- Automated gait recognition to identify individuals based on body shape and walking styles.
FER, University of Zagreb, Zagreb, Croatia (Sep 2018 – Dec 2019)
- De-Identification Using Deep Learning:** To modify geometry & texture, preserving dataset privacy without compromising naturalness.

JUL 2016 – APR 2024

4. Manipal University - Assistant Professor

JUL 2013 – JUL 2014

5. Sconad Communication, Mumbai, India Level 3 Research Associate

- Identified procedural areas of improvement through customer data to improve the profitability of a nationwide retention program by 8%.

EDUCATION

- 2018 – 2023 **Doctor of Philosophy**
Computer Science and Engineering (AI)
Manipal University, India
- 2014 – 2016 **Master of Technology**
Computer Science and Engineering (AI)
Maharshi Dayanand University, Rohtak, India
- 2009 – 2013 **Bachelor of Technology**
Computer Science and Engineering
Maharshi Dayanand University, Rohtak, India

SKILLS

Programming:	Python (advanced), Java.
Data Manipulation:	SQL, NoSQL, CSV, JSON, PySpark.
Data Analysis & Visualization:	Pandas, NumPy, Matplotlib, Seaborn.
Frameworks:	Scikit-learn, TensorFlow, Keras, PyTorch, Llama, ChatGPT.
Statistical Techniques:	Regression, Classification, Clustering, LLM, Agents, Dimensionality Reduction, RAGs, NLP, Computer Vision.
Cloud Platforms:	Hugging Face Hub, Azure, GCP.
Deployment & DevOps:	Flask, Docker, CI/CD.
Hardware:	Jetson Nano, Raspberry Pi, Arduino, Banana Pi.

PATENTS - (6) & JOURNALS - (42)

- A Covariate-based Gait Recognition System and Method for Edge Analytics Using Optimized Deep Learning Pipeline. *Indian Patent, Status: Granted (202111034240) (2022).*
- A Parashar, Advancements in artificial intelligence for biometrics: a deep dive into model-based gait recognition techniques, *Engineering Applications of Artificial Intelligence, 2024(Q1) SCI, IF – 8.34.*
- A Parashar, et.al., Deep Learning Pipelines for Recognition of Gait Biometrics with Covariates - A Comprehensive Review. *Artificial Intelligence Review., 2023 (Q1) SCI, IF – 9.588.*

MORE

COURSES

- Data Science Fundamentals with Python, SQL IBM
- Deep Learning Specialization. Stanford University
- Machine Learning Specialization. Washington univ.
- Deepstream for Video Analytics on Jetson. Nvidia
- AI on Jetson Nano. Nvidia
- Fundamentals of Digital Marketing. Google